

Checklist: Building a Production-Ready RESTful API with Spring Boot

1. Plan Your API

- Define purpose and scope
- Identify resources and endpoints
- Design request/response formats

2. Set Up Spring Boot Project

- Use Spring Initializr
- Add necessary dependencies

3. Define Entity Classes

- Use JPA annotations
- Add validation annotations

4. Create Repository Layer

- Extend JpaRepository
- Define custom query methods

5. Implement Service Layer

- Encapsulate business logic
- Handle exceptions

6. Build Controller Layer

- Use @RestController
- Implement CRUD endpoints
- Use DTOs and @Valid

7. Add Exception Handling

- Use @RestControllerAdvice
- Handle validation and runtime exceptions

8. Add Validation

- Use javax/jakarta validation annotations

- Validate DTOs in controller

9. Secure the API

- Use Spring Security
- Implement JWT-based authentication

10. Document the API

- Use Springdoc OpenAPI
- Annotate endpoints with Swagger annotations

11. Write Unit and Integration Tests

- Use JUnit and Mockito
- Use MockMvc and Testcontainers

12. Configure Application Properties

- Set datasource and JPA settings

13. Enable Logging

- Use SLF4J and Logback
- Configure log levels

14. Handle Pagination and Sorting

- Use Pageable in repository and controller

15. Deploy the API

- Package with Maven
- Deploy to Docker/Kubernetes/Cloud

16. Monitor and Maintain

- Use Spring Boot Actuator
- Integrate with Prometheus/Grafana